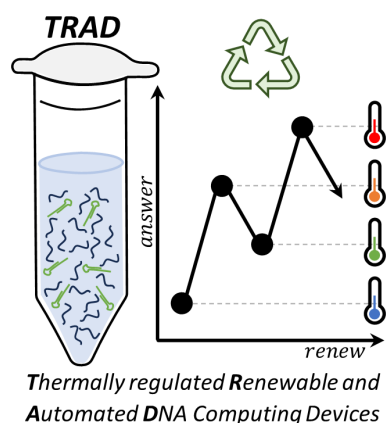


1 Quality, Significance, and Novelty

1.1 Introduction

Current digital data computation and storage technologies are hitting sustainability limits in terms of energy consumption and the use of rare and toxic materials ¹. Replacement materials are actively being researched. **DNA** is one of the promising materials because of its potential as a stable and robust data storage medium and its potential to carry out computations ^{2,3}. Also, producing **a system with zero waste** is a step forward in protecting the **environment**. A fundamental mechanism for most living organisms to adapt to their surroundings is their thermoregulatory system, which keeps the body healthy by increasing or decreasing its temperature depending on the surrounding environment and how the body responds to that change. Likewise, DNA hybridisation can be thermally regulated depending on the DNA's length, sequence, and structure, which permits applications like DNA Computing. With technological advancement, DNA computing devices with revolutionary designs solved several computation problems theoretically and/or experimentally ⁴⁻⁶.

However, DNA computers are typically **one-time use**, which is a **significant drawback**. Usually, after all the effort in designing and running the computation, the material in the test tube becomes waste despite the fantastic designs and their robustness ⁶⁻⁹. To run a new computation, one must at least carry out additional experimental steps to remix a new DNA program, which costs time, material, and labour. **The grand challenge of this proposal is to implement sustainable DNA programs that can be robustly reused not only repeatedly but also automatically.** The system proposed is renewable and self-contained, with no waste produced. By the end of the project, I expect to have a molecular DNA computer that can be pre-programmed and shipped to the user to run as many times as needed with a programmable temperature protocol based on the desired sequence of computations.



Most of the current DNA computing methodologies come with costs that affect the output with time. The output changes to an **error** due to problems like spurious nucleation in tile-based algorithmic self-assembly devices ⁶ or leaks in strand displacement devices ⁷. Such problems originate from the systems being **out of equilibrium** because the device output needs extra energy to maintain its correct state. **Thermodynamically Favoured Computation** (TFC) can address this because in a TFC system, DNA computations are designed where the output is the most energetically favoured structure, and no extra energy is needed to keep it in its state. Under-submission is a novel experimental design ¹⁰ called the Scaffolded DNA Computer (SDC). The SDC design is inspired by **the robustness and thermodynamic favourability** of DNA origami ¹¹, except that DNA origami does not compute. DNA origami is a technology utilising thermodynamic favourability to fold a long DNA strand; the scaffold; by - using short complementary strands or “staples” to form shapes in 2D or 3D at the nanoscale. In SDC, the output signal remains unchanged, and the run can be continued for a long time (tested for >24 hrs). That indicates the robustness of the SDC over time, which is longer than previously achievable reporting periods.

This proposal will address the drawback of the current DNA computing devices being a **one-time use**. The SDC showed preliminary results that it is renewable but not in an **automated way**. *To achieve sustainability and produce a zero-waste system*, I propose the novel idea of the **Thermally-regulated Renewable and Automated DNA Computer (TRAD)**. I can achieve that by integrating **thermal regulation** with **TFC** to produce a **one-pot** system that can **compute** and is **automatedly renewable**. The proposal has 4 main objectives addressing the grand challenge, and listed in the following table. Objectives are Specific, Measurable, Achievable, Realistic, and Time-bound within 48 months.

#	Research Objective	Work Package
O1	Validate a DNA Computing Platform with Thermodynamically Favoured Outputs.	WP1
	KPIs: D1.1 literature review (m6), D1.3 temperature programming model (m24),	
O2	Develop a Library of Thermally Regulatable DNA Motifs.	WP1
	KPIs: D1.2 library of 8 motifs (m18), D1.4 theoretical paper submission (m24)	
O3	Create an Automated One-Pot System for Molecular Computing.	WP2
	KPIs: D2.1 Experimental paper submission (m36), D2.2 TRAD prototype (m36)	
O4	Enable Energy-Efficient and Sustainable Molecular Computing Applications.	WP3
	KPIs: D3.1 Experimental paper submission (m48), D3.2 motifs DNA sequence large dataset (m48)	

All four objectives are tied to the deliverables of publishing in high-impact journals and conferences, recruiting, training, and mentoring a PhD student, and maximising the potential of DNA computing devices. **O1** is underpinned by the preliminary results of the SDC robustness and renewability. **O2** will take inspiration from previous elementary attempts at thermal cycling DNA devices¹² and design hairpin motifs¹³ that each can be activated at a unique temperature and are suitable to be inputs of the SDC. **O3** will integrate the input motifs with the selected thermo-regulatable device to achieve an automated one-pot renewable computing system. **O4** will validate the potential of the thermally regulated motifs to work on other thermodynamically favoured DNA computing platforms.

1.2 Background Context and current state of the art

One-time use of DNA computers. A significant drawback of most molecular computers, illustrated in **Figure 1**, is that the material used becomes waste after all the effort in defining models, designing sequences, and implementing programs⁶⁻⁹. DNA itself is cheap with the advancement of synthesising technology, but experimental material does not only contain DNA. It can have DNA and other chemical and biological elements that can be expensive, like the reporting molecules. However, there are attempts to make them reusable. Several reusable DNA computing platforms have been developed using approaches such as forcing dehybridisation by activation of photo-regulated molecules^{14,15}, controlling pH^{16,17}, using magnetic beads^{18,19}, using enzymes²⁰⁻²², switching temperature¹², or adding extra DNA strands^{10,23-26}.

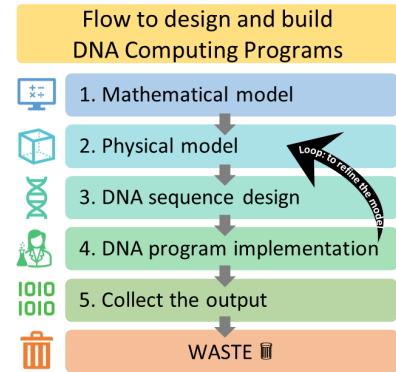


Figure 1: Typical flow of molecular computing programs.

Renewing with the same computation/input as before. Many of the approaches that achieve high yields renew/reset the program but run the exact same previous computation (input)^{12,16,17}. The thermal cycling approach¹² is a pure nucleic acid system (meaning no DNA modifications, enzymes, magnetic beads, or pH change) where the authors demonstrated how to include different motifs (3) in one solution and uniquely activate one at a specific temperature. The authors designed logic gates that can be reset by cycling the temperature, thus rerunning the program with the same set of inputs.

Renewing a program with a different input than before. Some designs²²⁻²⁶ run new computations by disabling previous inputs and introducing new ones. However, they tend to accumulate waste in solution. The more sophisticated the circuit is^{23,24}, the more complex and constrained renewal becomes, giving a poorer yield of the renewed cycles. Moreover, the waste from the previous computations severely interferes with the renewed computations and decreases the yield of the renewed outputs, indicating the circuit's inability to run more computations. The technique called clip-strand²⁶ allows for the changing of inputs and was used to build a full-adder logic circuit. However, each use and reset cycle needs about 12 hours to see the signal flatline, although signal levels leak over time. The SDC preliminary renewing results are promising. However, human interference is required to add the strands, and the liquid addition causes dilution in the program components, affecting the renewing signal¹⁰.

Overview of the Action. In summary, although the earliest DNA *motor* was renewable²⁷, designing renewable molecular *computers* has presented a significant challenge due to their complexity, out-of-equilibrium operation, and manual preparation steps. The principle of thermodynamic favourability of the SDC allowed for a simple and effective method for program renewal. However, automation of the renewing process is lacking. *I propose bridging this gap by developing automated renewable DNA programs that leverage thermodynamic-favourable computers and redesigning their inputs to be uniquely activated based on temperature.*

1.3 Novelty

A **one-pot** system means all device components are mixed together from the beginning, including every possible input. There is no need to prepare some device parts in advance and add the rest later as in previous DNA computing device methods^{2,6-9}. Automatically renewable means **eliminating human interference** to initiate a renewal cycle that can be a source of errors, which is the current state of the SDC and other renewable DNA designs^{10,23,24}. The vision of **TRAD** is to mix the program with all plausible inputs from the beginning and use pre-programmed thermal-regulation cycles to automate the running and renewing of the desired sequence of computations. There is no need for extra material, extra labour, extra cost, or external modifications like pH^{16,17}, UV^{14,15}, magnetic beads^{18,19}, enzymes^{20-22,28}, or even non-modified extra DNA strands addition²³⁻²⁶.

2 Methodology and Approach

2.1 Thermodynamically Favoured Computation, TFC

In a TFC, the resulting output is the most energetically favoured, meaning it has the least free energy than other structures and can maintain its state without needing external energy enforcement²⁹. In this context, SDC is the first case study to execute TRAD. In SDC, a single computation is run in one test tube. The components of a program are the scaffold, computing strands, and the input, **Figure 2.a** and **b**. A simple melt/anneal temperature protocol is used. Notably, the device will faultily compute various output structures in the absence of the input. Starting at a high temperature of 80°C, everything in the test tube becomes single-stranded, floating freely in the solution. Gradual temperature decrease helps the input drive the computation and avoid kinetic traps. At the end, the desired output structure of the computation forms. The rest of the device components remain in the solution without affecting the result. The melt/anneal result is measured using a fluorophore/quencher pair attached to reporting strands, and the signal intensity can be measured using a qPCR machine (or a plate reader). It was observed that the signal reaches its completion level at 53°C. Dropping the temperature afterwards does not affect the result. The stability of the resulting structures was tested by observing the output signal level at different temperatures held within the window of 50°C to 20°C, **Figure 2.c**. The signal remained flat at either a high or low reporting level for more than 10 hours. Moreover, the signal completion level is similar in all holding temperatures, which indicates that holding at any temperature below 50° C will not influence the result reporting.

To renew and reuse the device at the current SDC design¹⁰, (i) the old input is locked by adding its complement strand, (ii) the new input strand is added, and finally, (iii) the temperature protocol is reset. The device's new run is the new computation. New strand addition is done manually by pausing the experiment and manually pipetting the required strands' solution into the test tube. This process has **two implications**. **The first** is the time cost of stopping the experiment and the risk of human error. **The second** is the dilution of device components' concentrations due to the addition of liquid. Although computation was renewed more than ten times, a dilution effect was observed in the results (**Figure 2.d**). The figure shows results for the BIT-COPY computation in which the device acts as wire, either copying 0 or 1. The computation starts by copying 0 and then interchanges using the abovementioned procedure to alternate between Copy 1 and 0 in every cycle. With these preliminary results, I underpin **O1: Validate a DNA Computing Platform with Thermodynamically Favoured Outputs**, with more to explore after redesigning the inputs, as explained in WP1.

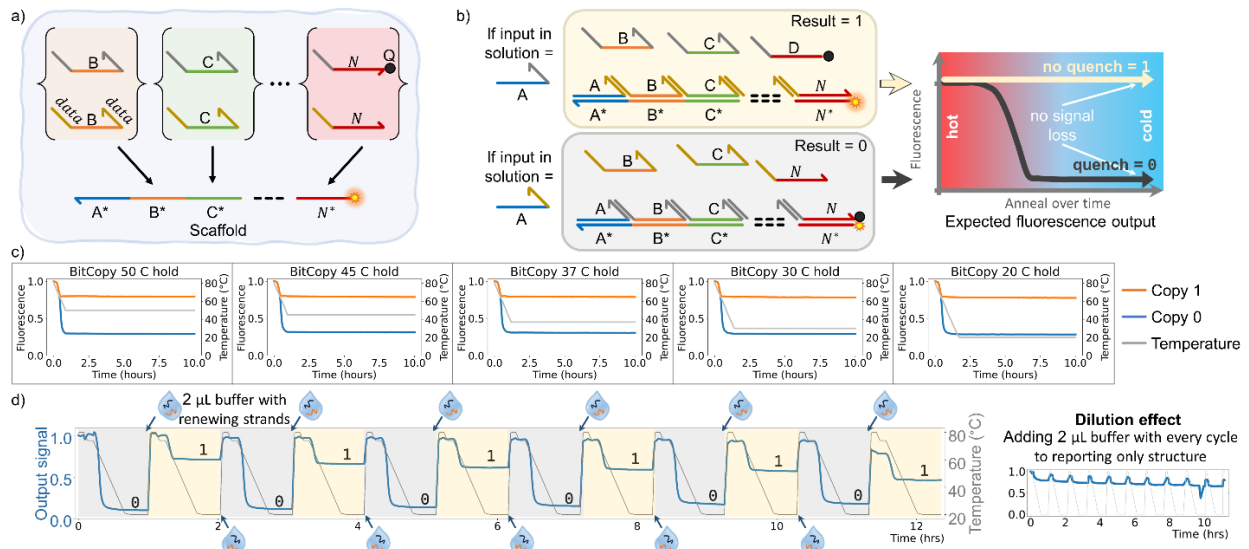


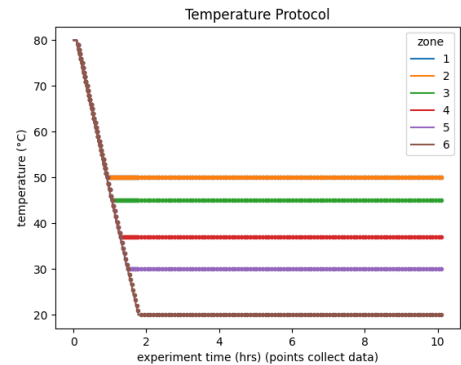
Figure 2: Scaffolding DNA Computer (SDC) and preliminary results. **a)** SDC's basic design shows competing computing strands over the scaffold. **b)** Expected computation structures and results in SDC. **c)** Preliminary bulk-fluorescence data showing stability and robustness of SDC BIT-COPY in different temperature holds. **d)** Preliminary bulk-fluorescence data showing a renewable SDC program. BIT-COPY computation alternates 12 times between Copy 0 and Copy 1, manually adding 2 μL of solution at the end of every computing cycle containing renewing strands. The right insert shows the dilution effect on signal reporting without computation.

2.2 Automation and data collection

To achieve *TRAD*'s automation, human interaction to renew needs to be eliminated. To achieve that, I see two parameters in the experimental design that must be addressed.

1. The control of the PCR machine temperature protocol must be automated.
2. The need to lock old inputs by hand-pipetting complements must be removed.

The first point will be addressed by pre-programming the sequence of computations' temperature protocols. I will use a software library developed in the Woods' group by Dr. Constantine Evans called *QSLib*³⁰, which allows online programmatic control (via Python) and feedback during experimental runs. *QSLib* has many useful features, like allowing repetitive runs, resetting the temperature protocol as often as needed, controlling the machine remotely and many other features. The qPCR machine currently in use is the QuantStudio™ 5. It can hold 96 tubes simultaneously. The heat block is divided into *six* zones with sixteen tube holders each. An example of a temperature protocol of the experiment in **Figure 2.c** is plotted on the right. In this experiment, the protocol started by holding all 6 zones for 5 minutes at 80 °C with no data collection. Afterwards, each zone's temperature was gradually dropped to a specific hold temperature value of [50°C, 50°C, 45°C, 37°C, 30°C, 20°C]. In the plot, zone 1 overlaps zone 2 because both get the exact temperature protocol and are held at 50°C.



QSLib code for this protocol is as follows:

```
channels = ['x4-m4']
newexp = Experiment(
    name= "Test",
    protocol = Protocol(
        stages = [
            Stage(Step(300, 80)),
            Stage(Step(time=100, temperature=80, temp_increment=-1, collect=True,
filters=channels, temp_incrementcycle=1), repeat=29),
            Stage.stepped_ramp(from_temperature=[50, 50, 50, 50, 50, 50],
to_temperature=[50, 50, 46, 46, 46, 46], total_time=8*60, n_steps=4, collect=True, filters=channels),
            Stage.stepped_ramp(from_temperature=[50, 50, 45, 45, 45, 45],
to_temperature=[50, 50, 45, 38, 38, 38], total_time=14*60, n_steps=7, collect=True, filters=channels),
            Stage.stepped_ramp(from_temperature=[50, 50, 45, 37, 37, 37],
to_temperature=[50, 50, 45, 37, 31, 31], total_time=12*60, n_steps=6, collect=True, filters=channels),
            Stage.stepped_ramp(from_temperature=[50, 50, 45, 37, 30, 30],
to_temperature=[50, 50, 45, 37, 30, 21], total_time=18*60, n_steps=9, collect=True, filters=channels),
            Stage(Step(time=300, temperature=[50, 50, 45, 37, 30, 20], collect=True, filters=channels),
repeat=100)
        ],
        volume = 35
    ),
    plate_setup=PlateSetup(setupFromGSsheet)
)
newexp.protocol.plot_protocol();
```

To address **the second point**, I propose redesigning the device's inputs to activate each at a specific temperature. DNA hairpins are an example of structures that can be regulated with temperature depending on their DNA sequence and the length of their stems and loops. A DNA hairpin structure happens when a single-stranded DNA, *ssDNA*, has complementary sites within its sequence. That makes the strand fold on itself, forming a hairpin shape, **Figure 3.a**. Hairpins can also happen with formations like having internal loops in their stems or bulges or leaving the beginning of one stem single-stranded and acting as a toehold. **Figure 3.b** shows a few examples of those formations. They can be achieved with careful sequence design. DNA hairpins form without errors because their formation does not depend on diffusion, i.e. other strands in solution. This property is called "locality" because the complementary sites are on the same strand. Their forming and shapes can be regulated by controlling the length and sequence of specific sites in the *ssDNA*. Let a hairpin be represented as $H(l, m, n, k)$, **Figure 3.c**, where l is length of the loop, m is length of internal toehold, n is length of external toehold, and k is length of the stem. A simple parametric model¹³ can be formulated based on this formula to regulate the functionality of the hairpin. The strength and stability of the hairpin at different temperatures also vary by the sequence of bases in the *ssDNA*. Sequence design will be needed and emulated using tools like NUPACK³¹.

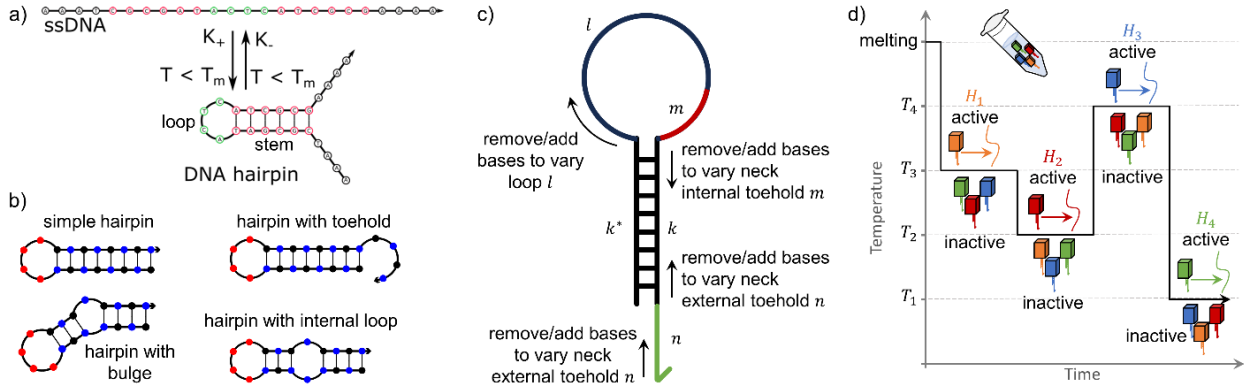


Figure 3: DNA hairpins. **a)** a DNA hairpin formation from a *ssDNA* when lower temperature T is lower than the melting temperature T_m . **b)** different formations of hairpin structures depending on the sequence of the DNA. **c)** modelling the functionality of the hairpin $H(l, m, n, k)$ by controlling lengths of specific sites in the *ssDNA*, l is length of the loop, m is length of internal toehold, n is length of external toehold, and k is length of the stem. **d)** A cartoon illustrating how at a distinct temperature one motif is activated (i.e. open and ready to react with the following strand in a circuit) while the others are inactive. Colours mean different hairpin formations but only one shape is used as a dummy structure.

Once modelled and designed, motifs will be experimentally tested using the following procedure:

1. All motifs will be mixed in equal concentrations in the same test tube. There will be a number of test tubes that match the number of the designed motifs, and all will have identical contents.
2. A fluorescence reporting mechanism will also be added to each tube.
3. Heat up all tubes to a high temperature to melt all secondary structures.
4. Drop the temperature of each tube to a specific temperature value that targets the activation of one motif while the others become inactive, as illustrated in **Figure 3.d**
5. Collect the bulk-fluorescence data for analysis.

The expected results of this experiment are that the activated motif will be the one reporting while the others are not. Different temperature protocols were tested on the SDC as preliminary work, and the results were consistent. One observation is that during the temperature protocol ramping down from 80°C to 20°C, the annealing reaction starts at ~72°C and finishes at ~53°C. The signal remains flat with a decrease in temperature afterwards. Another observation is that the reaction can still be held at different temperatures between the *working reaction temperature range* of <50°C to room temperature. Another observation was the quick response to temperature regulation when it dropped from 80°C to 55°C in 1 minute — well-separated signals between high and low signals. Therefore, the motif's unique activating temperature values can be specified within the *working reaction temperature range* of the targeted TFC, i.e. SDC in this case.

Once a thermally-regulated hairpin library is successfully implemented, objectives **O2: Develop a Library of Thermally Regulatable DNA Motifs**, **O3: Create an Automated One-Pot System for Molecular Computing**, and **O4: Enable Energy-Efficient and Sustainable Molecular Computing Applications** will be feasible. The goal is to generalise and maximise the uptake of the designed motifs library to be exploited and utilised by other DNA computing approaches that fall under the thermodynamically favourable computing category.

In summary, since **preliminary results** of the SDC and its renewability characteristic showed that it can be thermally regulated and by redesigning possible inputs of the device to be also thermally regulated, *this proposal will enable automation by having one pot that includes the device with all possible different inputs, each of which leads to a unique computation.* To run a specific computation, all that is needed is to choose the suitable temperature where, at that temperature, one specific input is activated while the others remain inactive, and the activated input leads the output structure to be the most energetically favoured at that temperature. The temperature cycles can be pre-programmed throughout the experiment using *QSLib* to control the real-time quantitative qPCR machines. No human intervention is needed. By that, the new design will **eliminate** the concerns of **human error**, **dilution**, and **accumulating waste** that interferes with the reaction and set the stage for a highly novel, renewable, programmable, and automated DNA computation system.

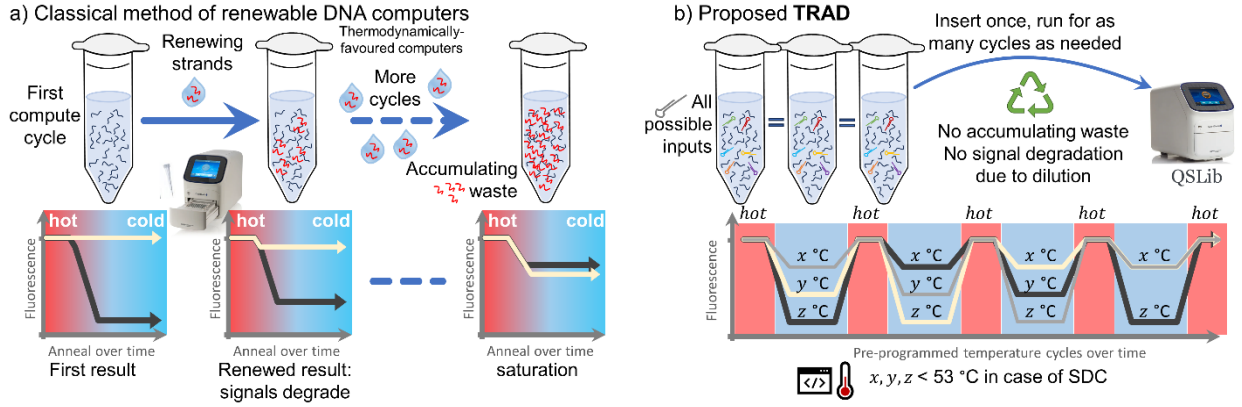


Figure 4: Project target. **a)** The typical approach to renew SDC as a thermodynamically favoured computer. First run result is the best that the program can accomplish either high or low. To renew the program, a human need to open the machine and pipette the renewing strands then reset the temperature protocol or run the next computing cycle. With a greater number of renewals over time, the signal degrades which can be a result of dilution and saturation because of the accumulating waste in the solution. **b)** TRAD. Identical tubes in content. Each of them contains the program and all possible inputs. Insert the tubes in the qPCR machine and let the automated protocol take place by running preprogrammed temperature cycles where each cycle holds at a temperature that activates only one input. No expected signal degradation as there is no accumulating waste or dilution due to liquid pipetting.

3 Work Plan

The work plan consists of 5 work packages (WPs) that address the grand challenge by addressing the project's four objectives.

WP#	Work Package Title	Associated Objective	Start month	End month	Lead
WP1	Theoretical Development and Design.	O1, O2	1	24	PI
WP2	Experimental Setup and Protocol Development.	O3	25	36	PhD
WP3	Sustainable Molecular Computing	O4	36	48	PhD & PI
WP4	Dissemination and Communication.	All Os	7	48	PI
WP5	Management and Training.	All Os	1	48	PI

3.1 WP1: Theoretical Development and Design (months 1-24, led by the PI)

Work Package associated objectives:

O1. Validate a DNA Computing Platform with Thermodynamically Favoured Outputs.

As illustrated in **Figure 4.a**, most previous renewable DNA systems that are pure DNA^{10,23-26}, including SDC, add strands to each cycle to achieve renewal. This proposal's expected setup and output are illustrated in **Figure 4.b**, which is a **one-pot** system that can **compute** and is **automatedly renewable**. As the proposal targets the TFC systems, **WP1** will start by surveying the literature on thermodynamically favoured DNA computers. SDC has shown that it is renewable by nature and is a pioneer in DNA TFC. Nevertheless, other researchers are expected to be interested in exploiting the reversibility of DNA thermodynamic computations²⁹. Also, since literature reviews pave the way for future researchers, a comprehensive review of renewable DNA computing with any approach or methodology will be drafted.

Meanwhile, as detailed in section 2.1, SDC's preliminary results showed that it is naturally renewable and can be held at different temperatures without affecting the resulting structure. **O1** will explore and verify that the selected temperature profiles of the SDC are compatible with the temperature profiles required by the hairpin motifs that will be designed under **O2**. There will be going back and forth between **O1** and **O2**.

O2. Develop a Library of Thermally-Regulatable DNA Motifs.

Given basic DNA hairpin $H(l, m, n)$, described in section 2.2, the goal is to design and test a set of hairpins that can act as inputs where each is activable at a specific temperature. Modulating the

functionality of a hairpin can be achieved by regulating specific sites in the *ssDNA*. Preliminary results of the SDC showed that it can function at various temperature holds within a range below its melting temperature. This objective will use that temperature range to model a set of hairpin structures, each of which can be activated at a specific temperature within that range. This objective will start immediately from the project starting month in parallel to O1. I expect at least twelve months of modelling to get distinct activation points. The higher the number of designed motifs, the more complex the sequence design will get. The first milestone will be 8 distinct motifs, which is a possible number of SDC inputs to encode 3 binary bits on the computing domain, which is the right-side domain of the input *A* in **Figure 2.b**. 3 binary bits gives the possibility of $2^3 = 8$ possible domains, hence possible inputs. I will work with the PhD student on the modelling problem. I aim to develop a general library under NUPACK designated for hairpin modelling with different formations, sequences, and sizes.

This work package will stimulate the design of other thermodynamically favoured computers, scaffolded or not. It will be an introductory learning phase for the student and an expansion of the expertise of the PI.

Deliverables: **D1.1** Literature review paper in an open science journal to allow open access for future researchers interested in the topic. Manuscript submission #1 is a literature review on renewable DNA computation methods to date. (*month 6*). **D1.2** A library of 8 thermally regulatable DNA hairpin-based motifs. (*month 18*). **D1.3** TFC temperature programming model. (*month 24*). **D1.4** Manuscript submission #2 is a theory paper accompanied by a modelling library in a field-specific annual DNA conference, like the annual International Conference on DNA Computing and Molecular Programming (*month 24*).

3.2 WP2: Experimental Setup and Protocol Development (months 25-36, led by the PhD)

Work Package associated Objectives:

O3. Create an Automated One-Pot System for Molecular Computing.

Once models have been developed and motifs are designed, DNA strands will be ordered to design and run experiments. Experiments will be divided into two sets. The first set of experiments will test and validate the designs of the hairpin motifs, as explained in the five-step procedure described in section 2.2. The second set of experiments will incorporate the input motifs with the existing SDC programs to test TRAD on SDC experimentally. The student will take the lead on this WP after being trained by the PI. Training will occur during the project's first two years in parallel with WP1. So, when month 24 is reached, the PhD student will be ready to lead by being fully trained on the wet lab procedures and protocols.

Deliverables: **D2.1** Manuscript submission #3 is an experimental paper in *Nature Communications* or a similar journal (*month 36*). **D2.2** A prototype of TRAD on SDC.

3.3 WP3: Sustainable Molecular Computing (months 36-48, PhD leads modelling, and PI leads experiments)

Work Package associated Objectives:

O4. Enable Energy-Efficient and Sustainable Molecular Computing Applications.

This work package is dedicated to expanding the use of thermally-regulated motifs to work with other thermally-favoured DNA computing designs. A more extensive library of motifs will be developed to encourage other researchers to adopt the approach. Using our model developed in WP1, we will aim to produce more motifs with distinct thermal regulations to satisfy the needs of various designs. In this work package, the PI will lead the experiments while the PhD student will lead the modelling and coding to expand and produce more motif designs. This workload division will allow the student to learn and be involved in all aspects of the project and graduate with a unique set of skills to accelerate their career in either academia or industry.

Deliverables: **D3.1** Manuscript submission #4 is an experimental paper in *Science*, *Nature*, *PNAS* or similar (*month 48*) **D3.2** A dataset of DNA sequences of thermally regulatable motifs (*month 48*).

3.4 WP4: Dissemination and Communication (months 7-48, led by the PI)

This WP's primary targets are 1) to ensure clear and effective communication about the project and its results to non-specialist audiences and 2) to ensure high-quality dissemination and further exploitation of scientific results to STEM and SSH professionals in academia, industry and government.

The findings will be disseminated using different approaches to communicate the project results effectively to relevant stakeholders and interested parties. The dissemination strategy will encompass various platforms, including but not limited to scholarly journals, webinars, seminars, colloquia, conferences, and talks. This comprehensive approach will facilitate the widespread distribution of research outcomes and encourage dialogue among experts and practitioners in the field.

Moreover, sex and gender are relevant in the Dissemination phase of the project. Therefore, I will ensure the results are disseminated and communicated by both males and females and to both males and females. Project results will be communicated in non-technical general interest media, which are expected to reach all genders equally. I aim to ensure no gender bias throughout, engaging relevant stakeholders gender-inclusively and using gender-neutral language.

Deliverables: **D4.1** Invited talk to present TRAD in field-relevant DNA conference *DNAx*, *FNANO* or similar. **D4.2** Publication of manuscripts from WP1,2, and 3. **D4.3** PhD thesis. **D4.5** Article to the public. **D4.6** RTE Brainstorm or similar

3.5 WP5: Management and Training (months 1-48, led by the PI)

This WP aims to polish the PI's project management skills and receive career development and non-scientific training. It will also develop the PhD student's research skills.

Deliverables: **D5.1** Career development plan. **D5.2** Data management plan. **D5.3** PhD student to participate in Maynooth University's Research Skills Development Programme (RSDP). **D5.4** Exploring job opportunities/invited talks. **D5.5** Further grant submission e.g. ERC Consolidator, to obtain funding to establish an independent research group.

Timing of work plan: Gantt Chart. Both PI and PhD will contribute to all deliverables. Only the lead is shown in the cells of deliverables.

Activity	Start month	End month	Year 1				Year 2				Year 3				Year 4			
			1	4	7	10	13	16	19	22	25	28	31	34	37	40	43	46
WP1	1	24																
D1.1	1	6	PhD	PhD														
D1.2	1	18	PI	PI	PhD	PhD	PhD	PhD										
D1.3	19	24							PhD	PhD								
D1.4	4	24		PI	PI	PI	PI	PI	PI	PI								
WP2	25	36																
D2.1	25	36									PhD	PhD	PhD	PhD				
D2.2	25	36									PI	PI	PI	PI				
WP3	36	48																
D3.1	36	48													PhD	PhD	PhD	PhD
D3.2	36	48													PI	PI	PI	PI
WP4	7	48																
WP5	1	48																

4 Conclusion and discussion about potential risks

The proposal considers the balance between proposing the exploitation of a fundamental approach and mitigating *risks*. Preliminary data in **Figure 2.b** and **c** show that a TFC-based DNA program is renewable and can work at a wide temperature range. WP1 will be successful when providing a number of thermally regulatable motifs to work as inputs to the current TFC case study, which is the SDC. This will mitigate the total failure of WP2 and WP3. In the worst case, WP3 will suffer from a limited number of produced motifs. However, this can be innovatively addressed by clever sequence designs and modelling of formations of the hairpin motifs.

Once thriving, this proposal will be a breakthrough in the DNA Computing field by building automated renewable DNA programs and making them reusable with no constraints. The main goal is to produce *a sustainable, renewable system with zero waste*. It is written with the target of being adopted by other research labs even in fields outside the DNA Computing field, like drug delivery, nanobiocircuits and sensors.

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